# A pipeline for comprehensive and systematic analyses of DNA methylation data

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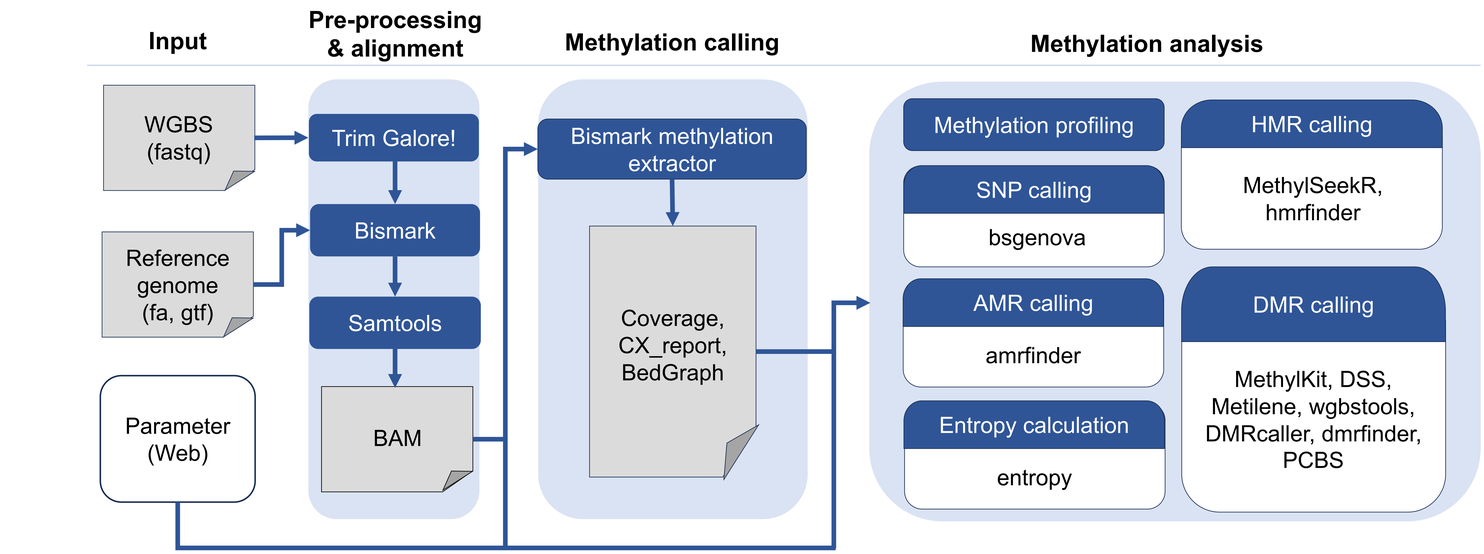
Intelligent Bioinformatics Computing Laboratory

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## Background

- DNA methylation is an important epigenetic modification that is involved in regulating gene expression, providing valuable insights into gene regulation, cellular identity, and disease mechanisms.
- However, many existing analysis tools require different complex configurations, diverse parameters, and non-standardized input formats, making them challenging for researchers unfamiliar with bioinformatics.
- To address these limitations, we developed a user-friendly end-to-end pipeline for analyzing DNA methylation data that supports an entire workflow from raw read preprocessing to comprehensive downstream analyses.

## Workflow and Features of pipeline



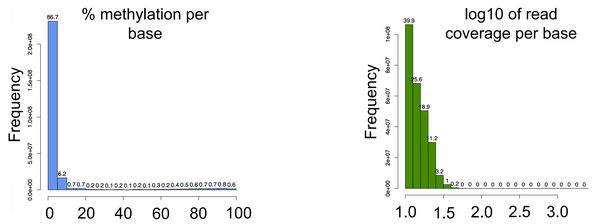

- Our pipeline was designed as a comprehensive pipeline to generate results of various kinds of DNA methylation analysis using raw whole-genome bisulfite sequencing(WGBS) reads.
- The pipeline follows a systematic approach beginning with whole-genome bisulfite sequencing read processing, proceeding through base-level methylation quantification, and extending to region-specific calling and analytical modules.
- It automatically generates genome-wide methylation profiles and provides comprehensive analytical outputs, including hypomethylated regions, differentially methylated regions, allelic-specific methylated regions, SNPs calling from methylation data, and methylation entropy.
- Beyond core quantification, our pipeline offers extensive downstream capabilities that enable biological interpretation, including genome-wide pattern summaries with PCA-based sample clustering, gene-level methylation characterization, various visualization plots for DMR-overlapping genes, and GO enrichment analysis of genes overlapping methylated regions.
- Leveraging containerization and parallelization technologies, our pipeline ensures a reproducible and efficient workflow across diverse computing environments. To further enhance accessibility, we implemented an intuitive web interface that allows users to easily configure key parameters through a guided approach.

## Acknowledgements

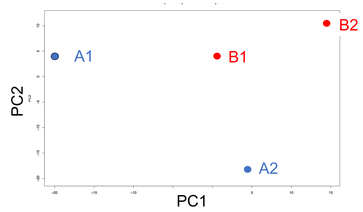
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## Conclusion

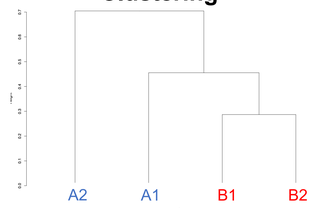
### General methylation statistics



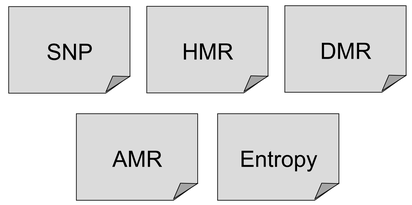

### PCA



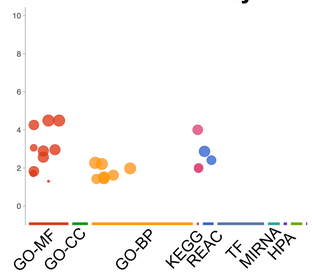

### Clustering




### Methylation calling




### GO enrichment analysis




- By significantly lowering the barrier to high-quality DNA methylation analysis, our tool enables broader participation in epigenetic studies.
- It facilitates the identification of methylation-based regulatory mechanisms and disease-associated epigenetic signatures in various human genome research.

## Reference

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